

Dr Moslem Uddin

Power Systems Engineer | Industry CV

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Engineering profile

Power systems researcher and engineer experienced in microgrid simulation, energy management, renewable integration and technical analysis.

Professional experience

Research Engineer / Research Associate, UNSW Canberra, 2025-present: hybrid microgrid modelling and energy management.

Research Assistant, UNSW Canberra, March-August 2025: simulation, validation and analysis.

Tutor and Lab Demonstrator, Universiti Teknologi PETRONAS, July 2016-February 2019: power flow and fault analysis laboratories.

Key projects

Advanced Energy Management Strategy for Community Microgrid, UNSW: hybrid solar, wind, battery and generation systems; multi-objective cost, emissions and reliability analysis.

Peak Load Shaving Strategies for Islanded Microgrid, Universiti Teknologi PETRONAS: battery-based load management and techno-economic evaluation.

Technical capabilities

MATLAB/Simulink, Python, HOMER Pro, DIgSILENT PowerFactory, PSS/E, PSCAD and ETAP.

Power flow and dynamic simulation, energy dispatch, optimization, system reliability and technical reporting.

Education

PhD in Electrical Engineering (Power Systems), UNSW Canberra, 2025.

MSc by Research in Power Systems, Universiti Teknologi PETRONAS, 2019.

BSc in Electrical and Electronic Engineering, Prime University, 2014; Chancellor's Gold Medal.

Selected publications

Community microgrid techno-economic study, Renewable Energy, 2023.

Microgrids: A review, outstanding issues and future trends, Energy Strategy Reviews, 2023.

Islanded microgrid battery peak shaving algorithm, Energy, 2020.